

Lesson 20 Functions
Input - output $f(x) = x^2$ ← formula/rule

$$2 \mapsto 4$$
$$g(x, y) = x + y \leftarrow$$

Def 1 A function f from a set A to a set B , denoted $f: A \rightarrow B$, is a rule which assigns each $x \in A$ to exactly one element $f(x) \in B$.

$\left\{ \begin{array}{l} \bullet \text{ If } x \in A, \text{ then } f(x) \in B \\ \bullet \text{ If } x, y \in A \text{ and } x = y, \text{ then } f(x) = f(y) \end{array} \right.$
well-defined (not ambiguous)

Image of f : $\text{im } f = \{ f(x) : x \in A \} \subseteq B$

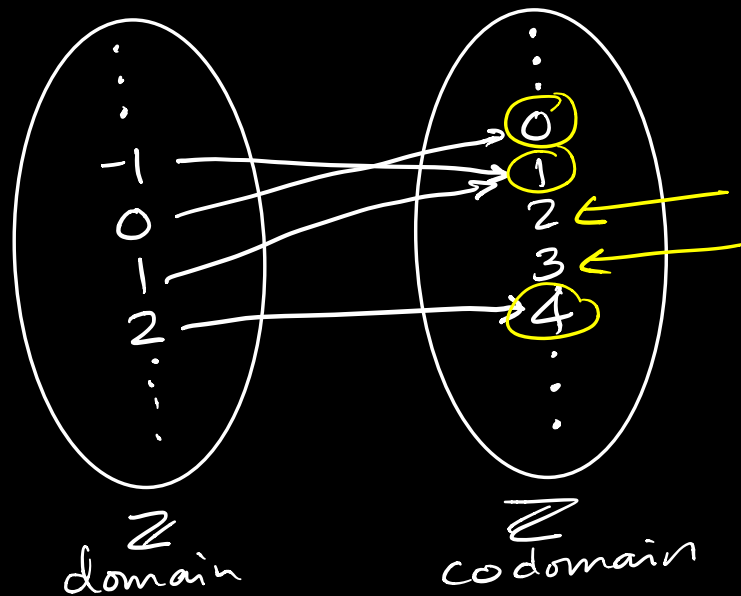
$f = g$: same rule, domain, codomain

$$f(x) = x^2$$

$$a) f: \mathbb{Z} \rightarrow \mathbb{Z}$$

a) $f(x) = x^2$ is a good rule
 $f: \mathbb{Z} \rightarrow \mathbb{Z}$ bc if $x \in \mathbb{Z}$, $x^2 \in \mathbb{Z}$
 $\uparrow$ exactly one

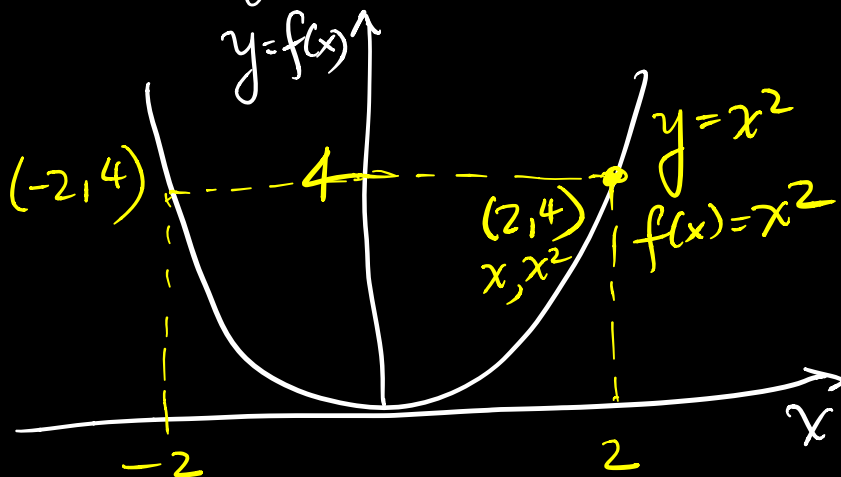
picture : ovals and arrows



$$\begin{aligned}
 \text{im} f &= \{0, 1, 4, 9, 16, \dots\} \\
 &= \{x^2 : x \in \mathbb{Z}\} \subseteq \mathbb{Z} \\
 &\quad \text{f(x)} \quad x \in A \quad \text{codomain}
 \end{aligned}$$

b) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ is good

picture : graph



$$\underline{\text{im}} f = \mathbb{R}_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$$

c) $f: \mathbb{Z} \rightarrow \mathbb{N}$
 $\quad \quad \quad \nwarrow$ positive integers

$f(x) = x^2$ is not good:

$0 \in \mathbb{Z}$ but $f(0) = 0^2 = 0 \notin \mathbb{N}$
domain codomain

not a valid fn.

d) $f(x)$ such that $f(x)^2 = x$

When $x=4$, $f(x) = \pm 2$
not well-defined

well-defined square root fn :

$$f: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$$

$$f(x) = \sqrt{x}$$

↖ non-negative root

pictures : arrows } $(x, f(x))$
graph }

Def 2 (equivalent/alternative def of fn)

A function f from set A to set B ,
 $f: A \rightarrow B$, is a relation from A to
 B , $f \subseteq A \times B$, such that:

- i) for every $x \in A$, $(x, y) \in f$ for some $y \in B$.
- ii) If $(x, y) \in f$ and $(x, z) \in f$, then $y = z$

every $x \in A$ is mapped to
exactly one $y \in B$.

$$(x, y) \in f \leftrightarrow y = f(x)$$

Def 2

Def 1

$$f: \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(x) = x^2 \quad (\text{Def 1})$$

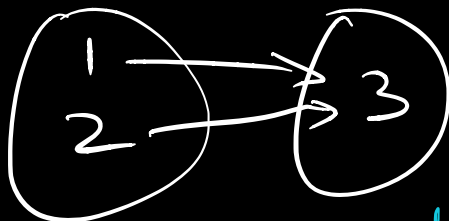
$$f = \{ \dots (-1, 1), (0, 0), (1, 1), (2, 4), \dots \} \subseteq \mathbb{Z} \times \mathbb{Z} \quad (\text{Def 2})$$

$$R = \{(1, 3), (2, 3)\}$$

R is a relation from $\{1, 2\}$ to $\{3\}$

R is from $\mathbb{Z}$ to $\mathbb{Z}$

R is a fn from $\{1, 2\}$ to $\{3\}$ ✓



It's ok to map 2 different elements to the same element
But not OK to map an element to 2 diff elements

R is NOT a fn from $\mathbb{Z}$ to $\mathbb{Z}$

$0 \in \mathbb{Z}$ but $(0, _) \notin R$
domain

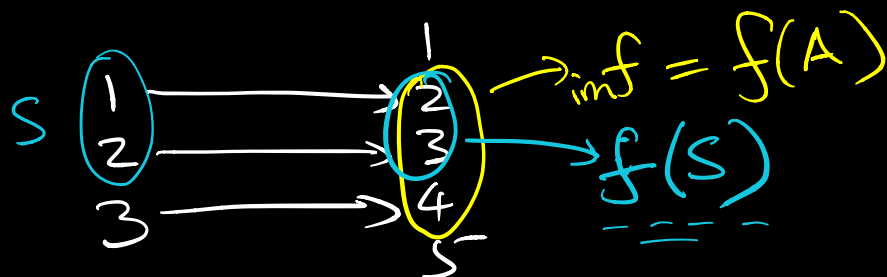
$f: A \rightarrow B$
image of f : $\{ f(x) : \underline{x \in A} \}$
whole domain

image of S (subset of domain) $S \subseteq A$

$$f(S) = \{ f(x) : x \in S \}$$

Ex $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$

$$f: A \rightarrow B, f(x) = x + 1$$



$$\underline{\text{im}f} = \{2, 3, 4\} = \underline{f(A)}$$

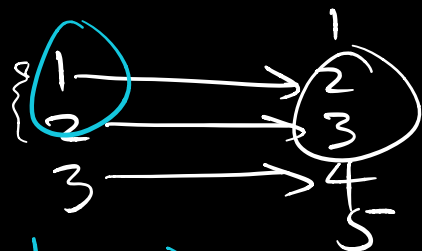
$$S = \{1, 2\} \subseteq A$$

$$f(S) = \{2, 3\}$$

Inverse image of $T \subseteq B$

$$T = \{2, 3\}$$

$$\underline{f^{-1}(T) = \{1, 2\}}$$



$$f^{-1}(B) = A$$

Exercises : prove

$$f(R \cup S) = f(R) \cup f(S)$$

$$f(R \cap S) \subseteq f(R) \cap f(S)$$

$$f(f^{-1}(T)) \subseteq T$$